



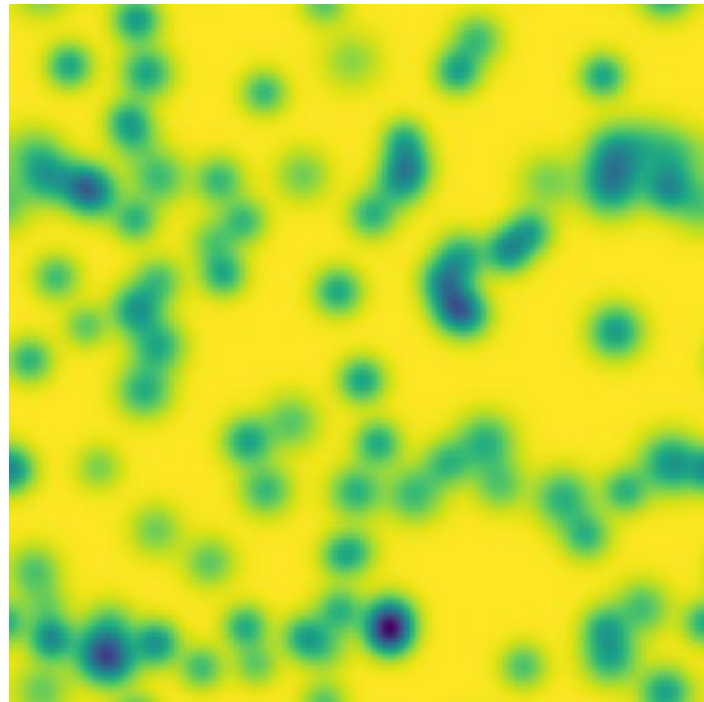
Latent Dynamics of Physical Fields

HackTheWorld — World Models

Team n° 18: JEPAdormi

Dataset – Physics-simulation

- Taken from The Well (Ohana et al., 2024)
- Trajectories of Gray-Scott reaction-diffusion systems
- 1,200 trajectories, 128x128 grid, 2 channels
- 154 GB of data



Example trajectory

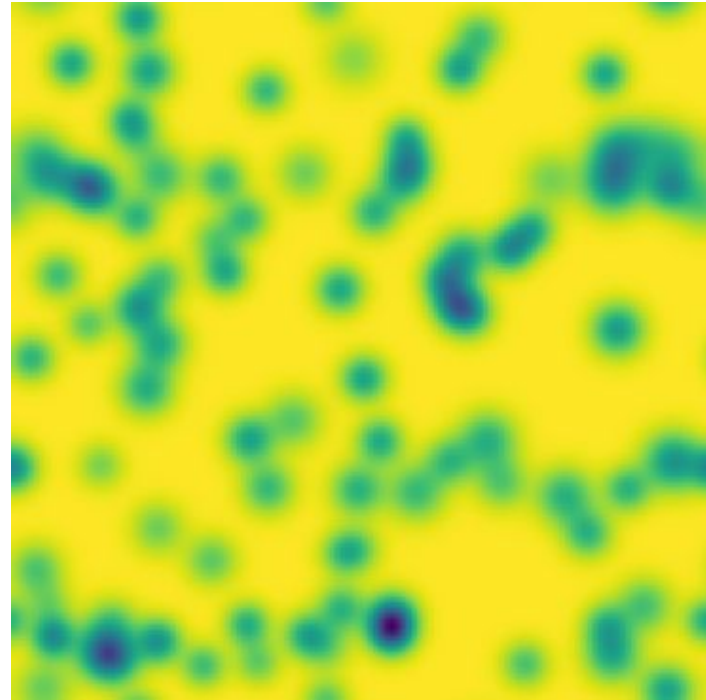
Dataset – Physics-simulation

Physics equation describing the system :

$$\frac{\partial A}{\partial t} = \delta_A \Delta A - AB^2 + f(1 - A)$$

$$\frac{\partial B}{\partial t} = \delta_B \Delta B - AB^2 - (f + k)B$$

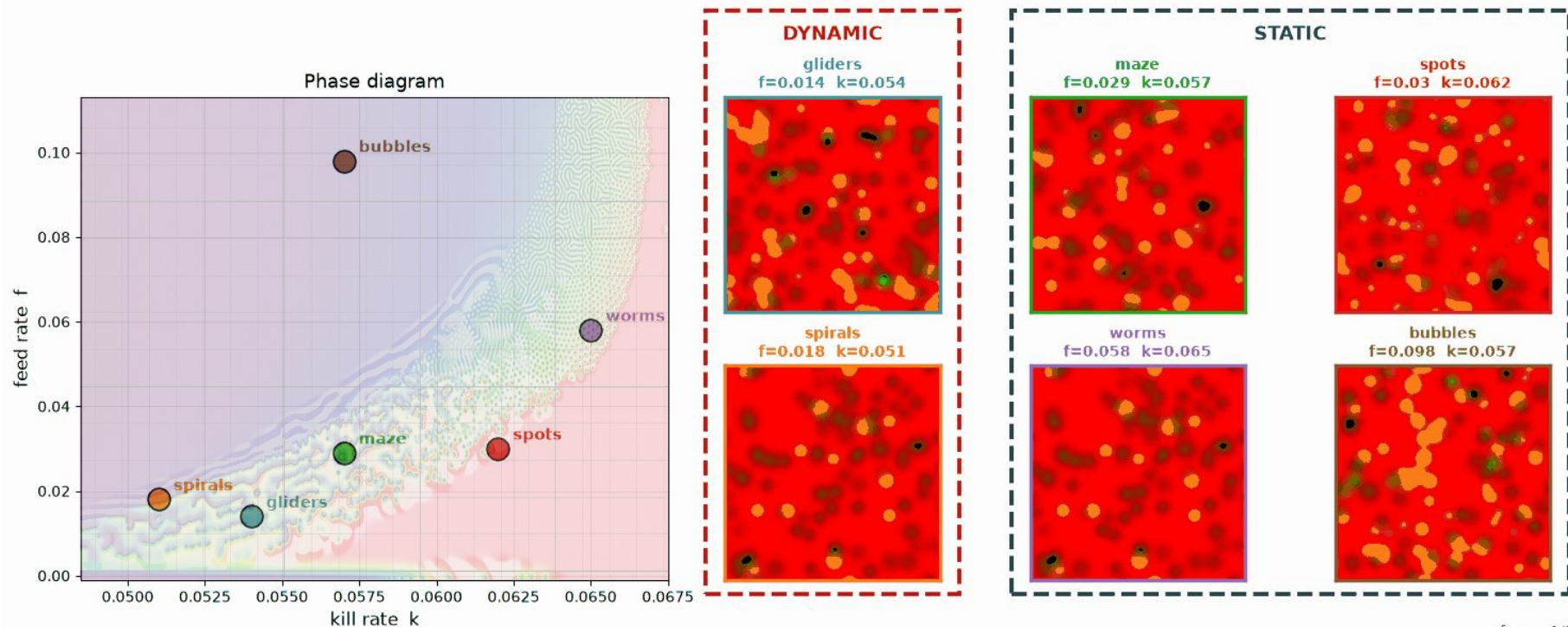
- $A(x,y,t)$, $B(x,y,t)$: concentrations of chemicals A, B at a given time and position
- f : feed rate
- k : kill rate



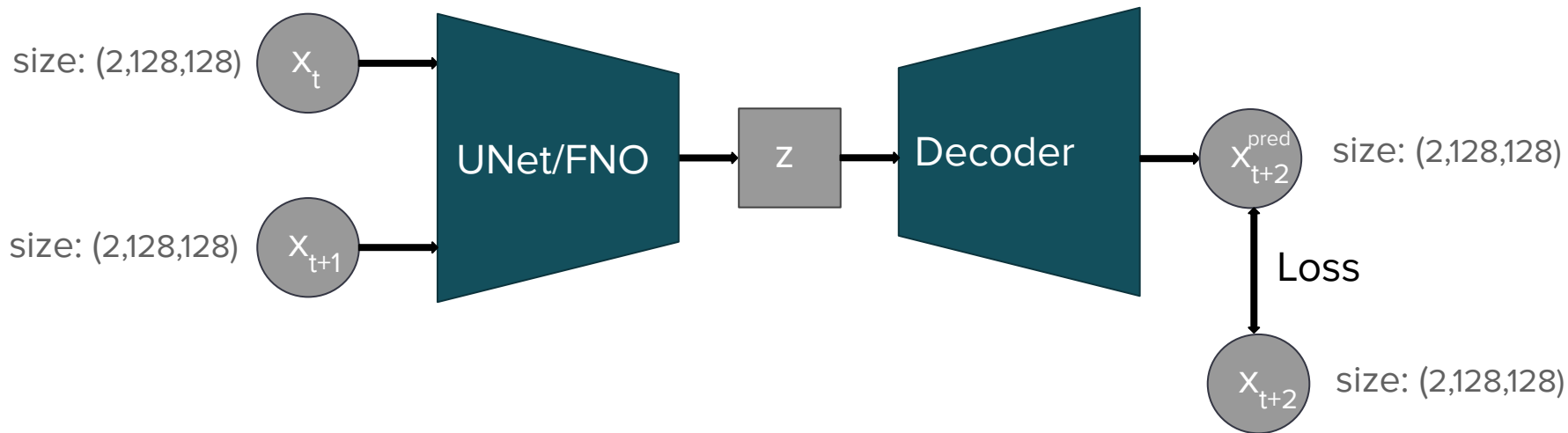
Example trajectory

Dataset – Physics-simulation

1200 trajectories $\rightarrow$ 6 distinct regimes (200 trajectories each)

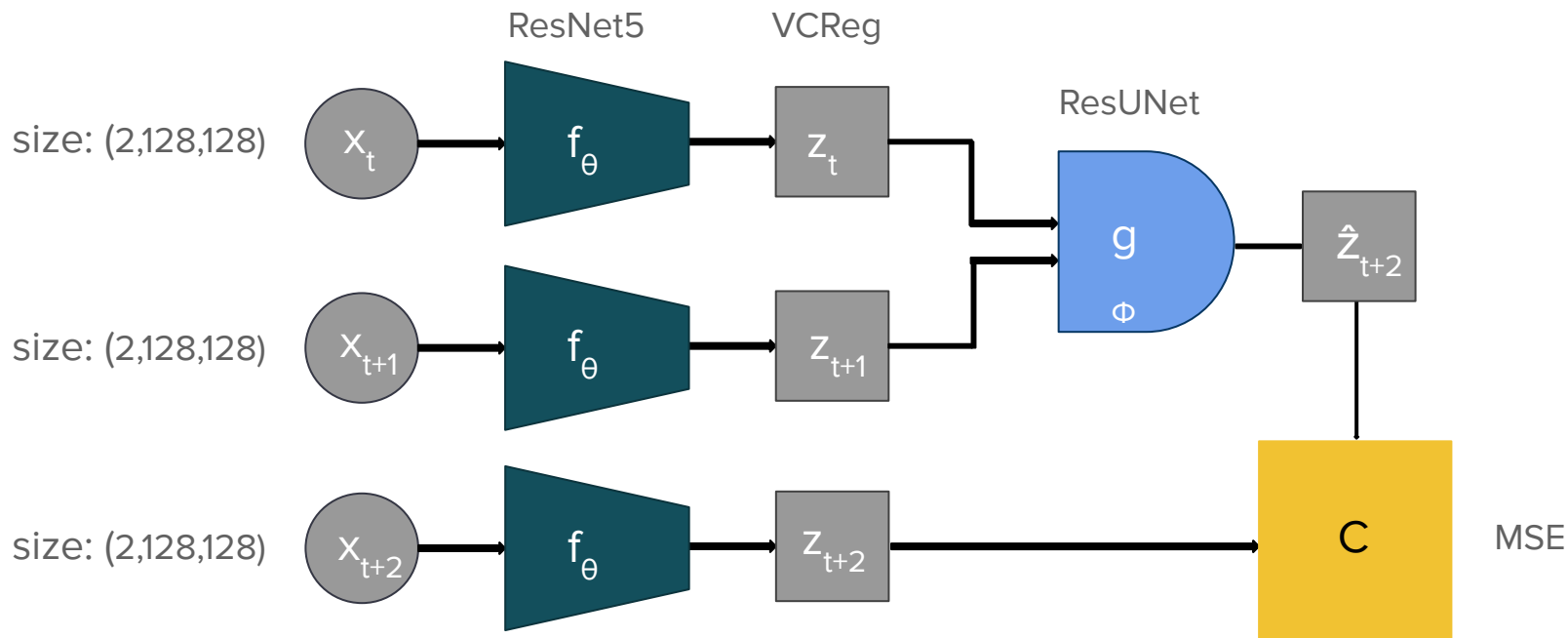


- Encoder-Decoder architecture using prediction in pixel space



Model architecture: JEPA

- Goal: learn latent physics
- Architecture: V-JEPA

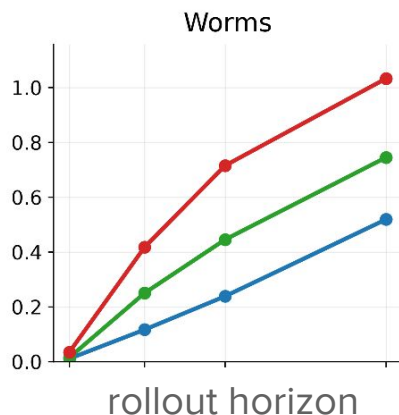
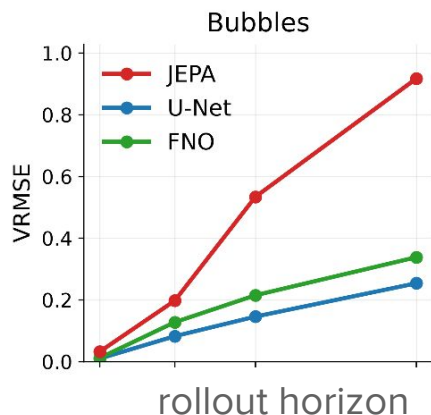


Metrics

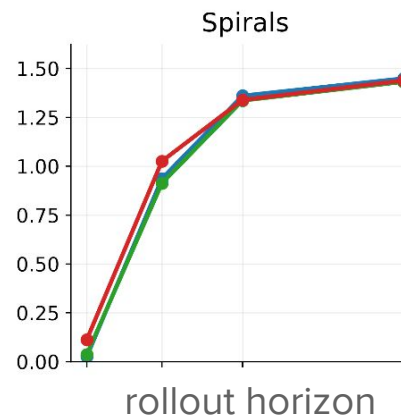
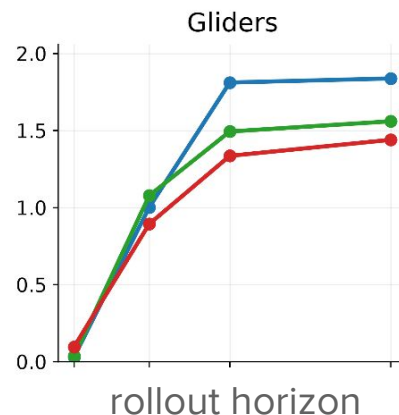


- Standard metric: VRMSE
$$\text{VRMSE}(x_{t+1}, \hat{x}_{t+1}) = \sqrt{\frac{\text{MSE}(x_{t+1}, \hat{x}_{t+1})}{\text{Var}(x_{t+1}) + \varepsilon}}$$

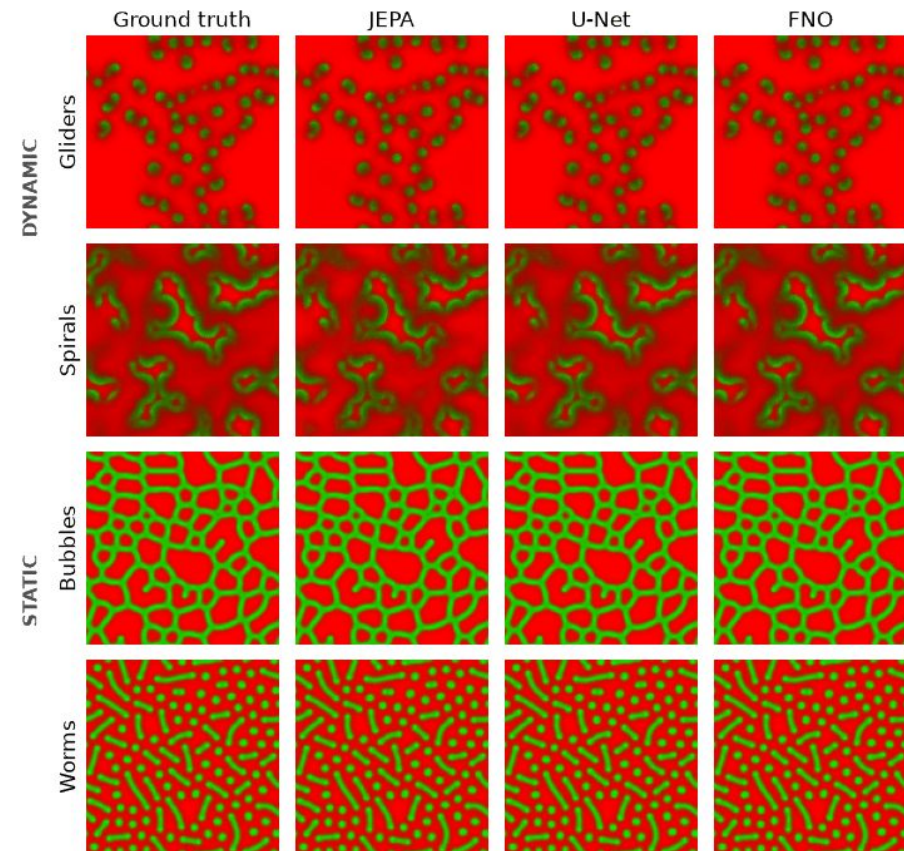
STATIC



DYNAMIC



Experimental Results

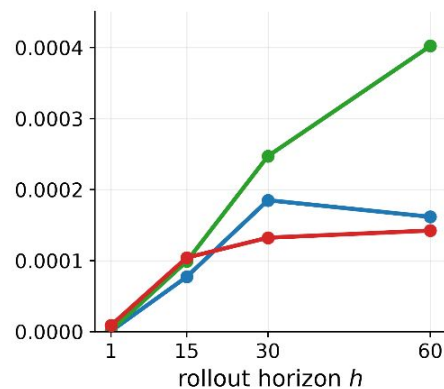
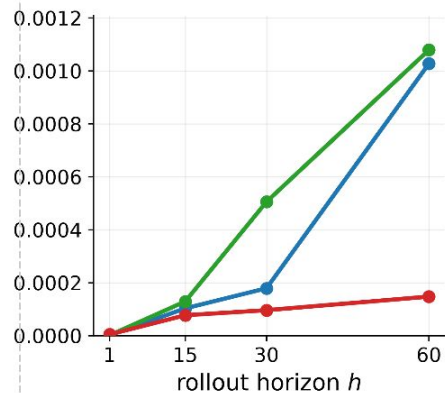
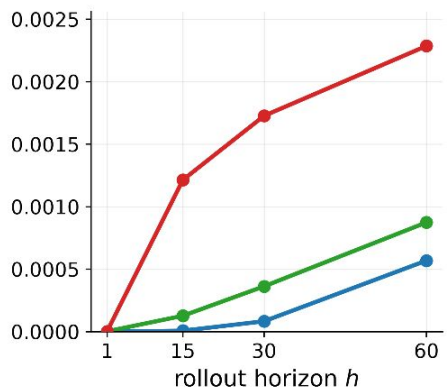
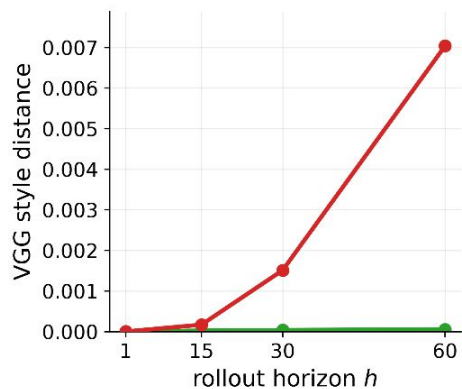


- JEPA learns to predict more realistic physics, especially on **long term horizons**
- Need for another metric

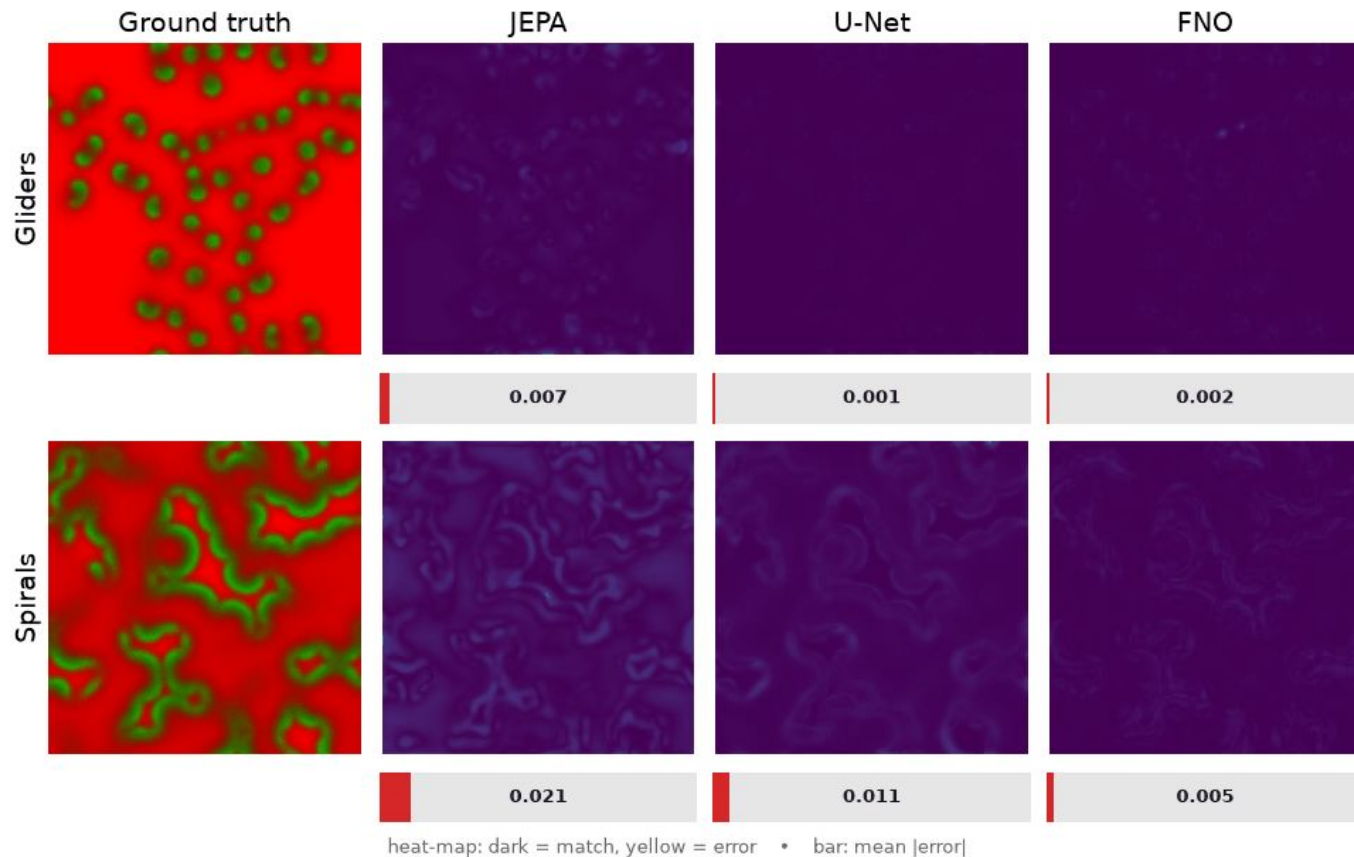
A new metric



- V-JEPA beats significantly the SOTA on dynamic phases
- Still struggles on quasi-static phase



Loss comparison with SOTA





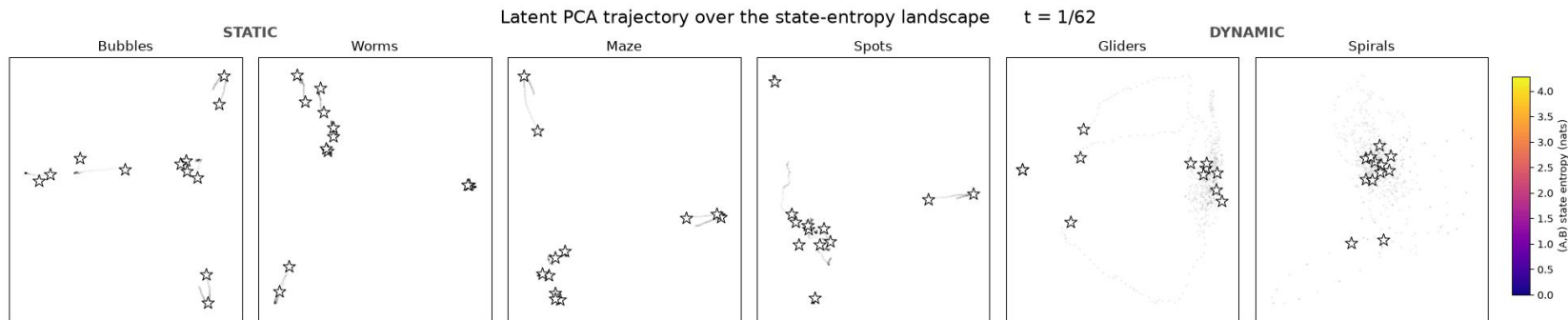
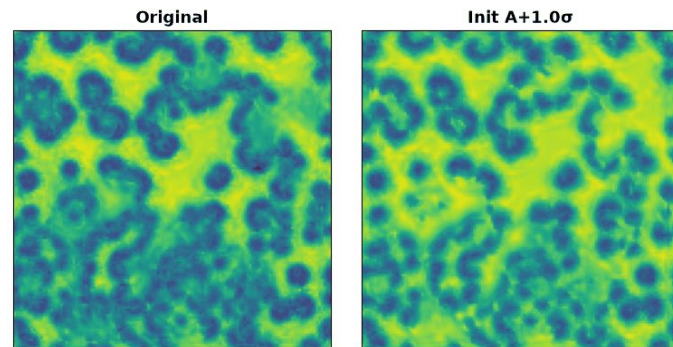
Recovered properties of the latent physics

PDE and Latent representations

The model does learn physical representations!

$$\frac{\partial B}{\partial t} = \delta_B \Delta B - AB^2 - \underline{\underline{(f + k)B}}$$

- **Latent PCA:** Pearson(4th component, $f+k$) = 0.7
- **Perturbations** of the initialization act according to the PDE
- **Entropie** of Shannon on a discrete histogram of (A,B)



Variance problem



$$\mathcal{L}_{\text{var}} = \frac{1}{D} \sum_{d=0}^D \max \left(0, \gamma - \sqrt{\text{Var}_t[z_{td}] + \varepsilon} \right) \quad \rightarrow \text{Var}_t[z_{td}] \geq \gamma^2 - \varepsilon$$

But

$(x_t)_t = \text{constante}$

$$\rightarrow \text{Var}_t[z_{td}] = 0$$

General Problem for JEPAs: Static Episodes (wait times, equilibrium...)

We need a different solution to Representation Collapse

Potential direction: correctors from e.g. fluid mechanics



Merci!
